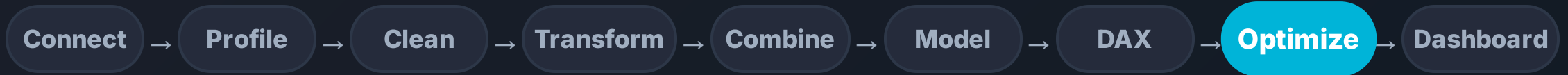


Optimizing Your Model

Hierarchies · Performance Analyzer · Column Removal

Where Are We?



Your model works. This week we make it better.

Week 16 → Presentations

Three Ways to Optimize



Hierarchies

Let users drill down — Year
→ Quarter → Month



Performance Analyzer

Measure how fast your
visuals load



Column Removal

Remove what you don't need
— leaner model

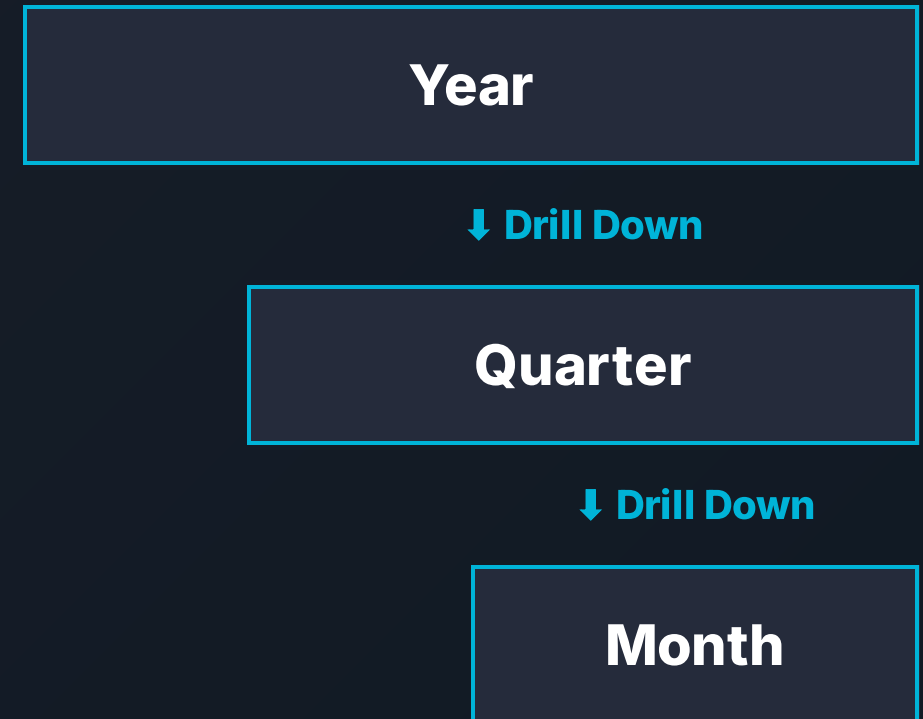
None of these change your data or your measures. They change how easy the model is to USE.

Hierarchies — The Staircase, Not the Elevator

"A hierarchy is a staircase, not an elevator. You walk from Year to Quarter to Month, one level at a time."

Without a hierarchy → flat list of dates.

With a hierarchy → users navigate from big picture to detail.



Building a Hierarchy — Three Drags

1. Model View → Date table
2. Right-click Year → Create Hierarchy
3. Drag Quarter onto the hierarchy
4. Drag Month Name onto the hierarchy

✓ **Date Hierarchy with 3 levels**

[Screenshot: Model View showing Date table with hierarchy created – Carlos to insert from his Power BI Desktop]

Drill-Down in Action



The double-arrow icon expands all bars one level. Click it twice to go from Year to Month.



Common Mistake: If months appear alphabetically (April, August, December...), you need **Sort by Column → Month Number**. You set this up in Week 12.

Performance Analyzer — Your Stopwatch

View tab → Performance Analyzer → Start Recording
→ Refresh Visuals

DAX Query	Time the engine spends calculating
Visual Display	Time the browser spends rendering
Other	Background processing

Every visual runs its own query. This tool tells you which one is slow.

[Screenshot: Performance Analyzer panel showing timing results — Carlos to insert]

On 500 rows, everything is fast. On 5 million rows, this tool saves you hours of guessing.

Column Removal — Less Is More

Before: 12 Columns

customer_id	name
email	phone
license_plate	address
city	state
zip	signup_date
loyalty_tier	expected_attendance

After: 8 Columns

customer_id	name
address	city
state	zip
signup_date	loyalty_tier 
email	phone
license_plate	expected_attendance



The Rule: If a column is not used in a relationship, a measure, a visual, or a filter — it probably does not belong in the model.

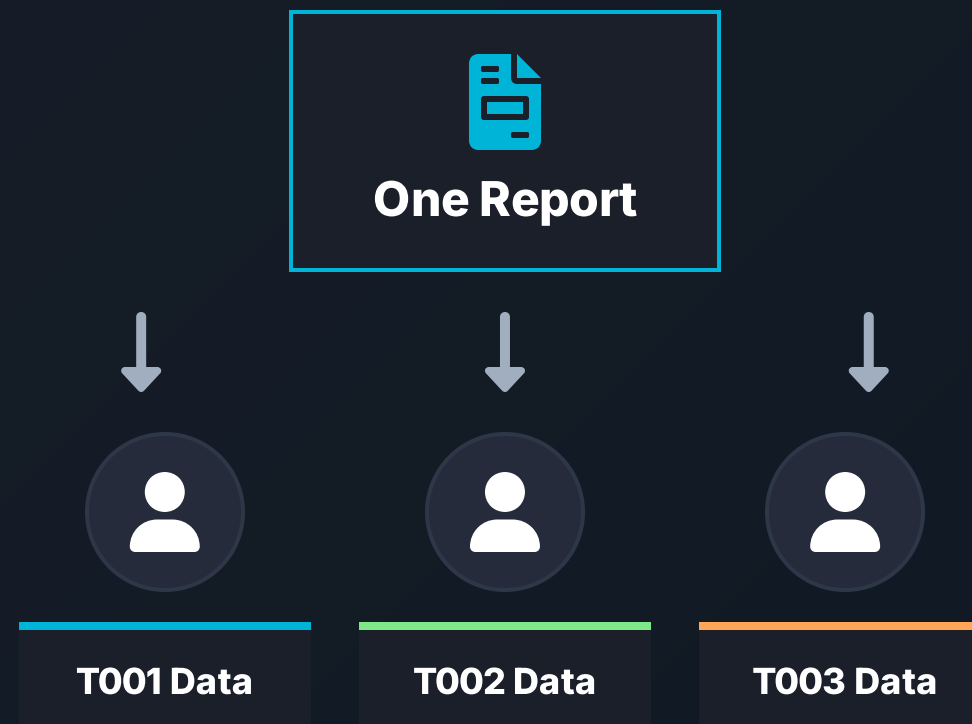
 **Hide in Report View** → safe, reversible  **Remove in Power Query** → permanent, more optimization

Row-Level Security — Locks on Specific Rooms

"Everyone enters the same house, but some rooms have locks. Your key determines which rooms you can enter."

Sabor Miami Example: Truck T001 manager sees only T001 sales. The filter goes on the Trucks dimension table, and the star schema carries it to Sales automatically.

⚠ Awareness only — not in Lab 4



Lab 4 — Wednesday, 50 Points



Date Hierarchy

8 pts



Performance Analyzer

10 pts



Column Removal

8 pts



Dashboard Page

16 pts



Written Response

8 pts



The dashboard page is 16 points — the biggest single item. Plan it. Sketch it on paper today during tutoring.

Pick Your Partner — Before You Leave Today



Step 1 — Wednesday Tutoring

Compare Lab 4 files, pick the stronger one, polish together



Step 2 — Weekend

Finalize the shared .pbix file



Step 3 — Monday Week 16

Present together, 5 minutes, both partners speak

Logistics

- ✓ Exchange contact info today
- ✓ Make sure you can communicate between now and Monday

No Partner?

Talk to your instructor before you leave today.

The Finish Line Is Visible

Today	Guided practice (hierarchy + Performance Analyzer + column removal) + partner selection
Monday tutoring	Continue practice, sketch your dashboard layout
Wednesday class	Lab 4 (individual, graded, 50 pts)
Wednesday tutoring	Portfolio clinic with your partner
Monday Week 16	Presentations (5 min per group)

✓ Hierarchies — drill-down navigation

✓ Performance Analyzer — find slow visuals

✓ Column Removal — lean model

Two more sessions of building. Then you present what you built.